

Real-Time Fraud Detection System

Overview

Financial fraud is becoming more sophisticated every day, and traditional manual monitoring is no longer enough to detect suspicious transactions quickly. To address this challenge, I developed a Real-Time Fraud Detection System that identifies fraudulent transactions within milliseconds and provides clear explanations for every prediction.

This project was built using the IEEE-CIS Fraud Detection on Dataset, which contains over 590,000 real-world financial transactions. One of the biggest challenges in the dataset was the severe class imbalance — only about 3.5% of the transactions were actually fraudulent. To solve this problem, I used SMOTE (Synthetic Minority Oversampling Technique) to generate realistic fraud examples and improve the model's ability to learn fraud patterns effectively.

Models Used

Two advanced machine learning models were trained and evaluated:

- LightGBM
- XGBoost

After comparing both models across several performance metrics, LightGBM produced the best overall results.

Final Performance of LightGBM

- ROC-AUC: 0.961
- PR-AUC: 0.784
- Fraud Recall: 87.3%
- F1-Score: 0.841

These results show that the model can successfully identify the majority of fraudulent transactions while minimising false alarms.

Risk Classification System

Instead of simply labelling transactions as “fraud” or “not fraud,” the system categorises them into clear risk levels:

Critical Risk

- Fraud probability greater than or equal to 0.75
- Transactions in this category are highly suspicious and require immediate investigation.

Suspicious

- Fraud probability between 0.40 and 0.74
- Transactions are flagged for manual review.

Clear

- Fraud probability below 0.40
- Transactions are considered safe with a very low chance of fraud.

This approach makes it easier for fraud analysts to prioritise their investigations efficiently.

Explainable AI with SHAP

One of the major limitations of many AI systems is the “black box” problem, where predictions cannot be easily explained. To overcome this issue, I integrated SHAP (Shapley Additive explanations) into the system.

SHAP helps explain why a transaction was flagged by showing which features contributed most to the fraud prediction.

Important Features Identified

- Large transactions on amounts
- Repeated card identifiers linked to fraud
- Unusual device or browser behaviour
- High transaction frequency within short periods
- Address mismatches and sudden spending spikes

This makes the model more transparent and trustworthy for auditors, compliance teams, and financial analysts.

Major Fraud Patterns Discovered

During analysis, the system uncovered several interesting fraud behaviours:

Night-Time Fraud Activity

Fraud rates were significantly higher between midnight and 3 AM, likely because monitoring is weaker during these hours.

Round-Number Transactions

Transactions with clean amounts such as \$100 or \$500 were more likely to be fraudulent, which is common in card transactions.

High-Value Spending Outliers

Fraudulent transactions were often much larger than a customer's normal spending behavior.

Recommended Prevention Policies

Based on the model's findings, several practical fraud prevention policies were suggested:

1. Hold high-value high-risk transactions for OTP verification.
2. Escalate suspicious round-number transactions for manual review.
3. Temporarily freeze cards showing unusually high transactions on velocity.

These strategies could potentially save financial institutions over \$1.2 million annually.

Limitations

Although the project achieved strong results, there are still some limitations:

- Fraud patterns continuously evolve and require regular retraining.
- Some identity-related features in the dataset were missing.
- Synthetic fraud samples may not capture every new fraud strategy.
- Threshold optimisation for real business environments can be further improved.

Future Improvements

The system can be enhanced further by including:

- Device fingerprinting
- IP geolocation on tracking
- Merchant risk scoring
- Transaction on velocity analysis

Conclusion

This project demonstrates how machine learning can be used to build an intelligent and explainable fraud detection system capable of identifying fraudulent financial activity in real-time. By combining strong predictive performance with explainable AI techniques, the system not only improves fraud detection accuracy but also builds trust with analysts and compliance teams.

Overall, the project highlights the practical impact of AI in financial security and showcases how data-driven systems can help organisations reduce fraud-related losses efficiently.